United States Patent Application Publication	20250273246
Kind Code	A1
Publication Date	August 28, 2025
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MAGNETIC DISK DEVICE

Abstract

A magnetic disk device includes a housing, a disk rotatably provided in the housing, a head provided in the housing and capable of writing and reading information to and from the disk, a voice coil motor provided in the housing and driving the head, and a filter structure provided in the housing and having a porous body including a first porous body with an average pore diameter of 0.4 nm or more and 1.0 nm or less, that is optimized for adsorbing glycol ether.

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Family ID: **1000008421753**

Appl. No.: **19/029506**

Filed: **January 17, 2025**

Foreign Application Priority Data

JP	2024-028380	Feb. 28, 2024

Publication Classification

Int. Cl.: **G11B33/14** (20060101)

U.S. Cl.:

CPC **G11B33/146** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-028380, filed Feb. 28, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a magnetic disk device.

BACKGROUND

[0003] A hard disk drive (HDD) includes a magnetic disk that spins at a high speed and a head for reading and writing information from and to the magnetic disk. The presence of large amounts of gas molecules in a housing of the HDD can deteriorate controllability of a distance between the head and the disk, which may shorten the lifetime of the HDD.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1** is a perspective view illustrating a magnetic disk device according to a first embodiment.

[0005] FIG. **2** is a schematic plan view illustrating the magnetic disk device according to the first embodiment.

[0006] FIG. **3** is a schematic plan view illustrating a modification of the magnetic disk device according to the first embodiment.

[0007] FIG. **4** is a top view illustrating an example of a structure of a first filter.

[0008] FIG. **5** is a perspective view illustrating an example of the structure of the first filter.

[0009] FIG. **6** is a schematic plan view illustrating an example of the structure of the first filter.

[0010] FIG. **7** is a schematic cross-sectional view illustrating a magnetic disk device according to a second embodiment.

[0011] FIG. **8** is a diagram illustrating an example of a structure of a second filter.

DETAILED DESCRIPTION

[0012] Embodiments provide a magnetic disk device with excellent performance in removing gas molecules having a predetermined valid diameter.

[0013] In general, according to one embodiment, the magnetic disk device includes a housing, a disk rotatably provided in the housing, a head provided in the housing and capable of writing and reading information to and from the disk, a voice coil motor provided in the housing and driving the head, and a filter structure provided in the housing and having a porous body including a first porous body with an average pore diameter of 0.4 nm or more and 1.0 nm or less, that is optimized for adsorbing glycol ether.

[0014] Embodiments of the present disclosure will be described below with reference to the drawings.

[0015] The drawings are schematic or conceptual, and the relationship between the thickness and width of each part, the ratio of the size between the parts, and the like are not necessarily the same as the actual ones. Even when the same parts are represented, the dimensions and ratios of the same parts may be represented differently from each other depending on the drawings.

[0016] In the present specification and each drawing, elements similar to those described earlier with respect to the previously described drawings are denoted with the same reference numerals, and detailed descriptions thereof will be omitted as appropriate.

[0017] In description, a direction from a base **11** to a top cover **12** of a housing **10** is referred to as “up” and a direction from the top cover **12** to the base **11** is referred to as “down”. However, the “up” and “down” directions are not limited to the direction of gravity or the direction at the time of implementation of the magnetic disk device.

First Embodiment

[0018] FIG. **1** is a perspective view illustrating a magnetic disk device **100** according to the present embodiment.

[0019] The magnetic disk device **100** includes a housing **10**, which includes a box-shaped base **11** and a top cover **12**. The base includes a bottom surface **11***b* and a side surface **11***w* that rises along the edge of the bottom surface **11***b* and is integrally formed with the bottom surface **11***b*. The housing **10** is sealed when the top cover **12** is combined with the base **11** by a plurality of joining parts. For example, the joining part may be a screw or the like, or an adhesive or a fit-in mechanism may be utilized. FIG. **1** illustrates an example of joining by a plurality of screws **13**. The material of the base **11** or the top cover **12** may be metal including, for example, Al or Fe.

[0020] The base **11** is provided with a disk **20** for recording data, a spindle motor **21**, and an actuator **30**. The spindle motor **21** rotatably holds the disk **20**. The actuator **30** includes a head **31** that scans the surface of the disk **20**, and reads and writes information. The actuator **30** further includes an arm **32** on which the head **31** is attached and a voice coil motor **33** that controls a position of the arm **32**.

[0021] The base **11** is provided with a substrate **40**. The substrate **40** is electrically connected to the actuator **30** via a flexible printed circuit (FPC) substrate **41**. The substrate **40** controls the movement of the head **31** and the voice coil motor **33**.

[0022] A plurality of disks **20** may be mounted on the spindle motor **21**. For example, two, three, or four disks **20** may be mounted. Alternatively, five or more disks **20** may be mounted.

[0023] On the base **11**, a first filter **51** is provided in the vicinity of the disk **20**. The first filter **51** is provided in a position in which at least a portion of the airflow generated by the rotation of the disk **20** passes through. The first filter **51** includes a porous body PM. Details of the porous body PM will be described later.

[0024] The top cover **12** may be provided with a hole **12***h*. A second filter **52** may be further provided in the housing **10** in a position corresponding to the hole **12***h*. The second filter **52** will be described later with reference to FIGS. **7** and **8**.

[0025] FIG. **2** is a plan view of the magnetic disk device **100** illustrated in FIG. **1**.

[0026] The base **11** is provided with the first filter **51**. FIG. **2** illustrates an example in which the base **11** has four corners. FIGS. **1** and **2** illustrate an example in which the first filter **51** is disposed between a first corner **11***c***1** of the base **11** and the disk **20**. A corner is a portion of the side surface **11***w* of the base **11** that has a curvature. The first corner **11***c***1** is one of the corners located opposite to the actuator **30** or the substrate **40** with respect to the disk **20**. In an X direction, the disk **20** is located between the first corner **11***c***1** and the actuator **30** or the substrate **40**.

[0027] The first filter **51** is provided beside the disk **20**. A distance between the first filter **51** and the disk **20** is shorter than a distance between the voice coil motor **33** and the disk **20**. The first filter **51** is provided opposite to the voice coil motor **33** relative to the disk **20**. In other words, the first filter **51** is located on an opposite side of the disk **20** relative to the voice coil motor **33**.

[0028] FIG. **3** illustrates an example of providing the first filter **51** between the disk **20** and the actuator **30** as a modification of the magnetic disk device **100** according to the first embodiment. Among the plurality of corners of the base **11**, the corner closest to the actuator **30** is referred to as a second corner **11***c***2**. FIG. **3** illustrates an example in which the first filter **51** is provided in the vicinity of the second corner **11***c***2**. Being in the vicinity of a certain corner refers to being closer to the certain corner than to other corners.

[0029] The first filter **51** may also be provided, for example, between the disk **20** and the substrate **40**. The first filter **51** may be provided in the vicinity of any of the plurality of corners of the base **11**. The first filter **51** is provided beside the disk **20**. A distance between the first filter **51** and the disk **20** is shorter than a distance between the voice coil motor **33** and the disk **20**.

[0030] In either case of FIGS. **2** and **3**, the first filter **51** is provided beside the disk **20**, and airflow generated around the disk **20** passes through at least a portion of the first filter **51**.

[0031] Subsequently, an example of the structure of the first filter **51** will be described with reference to FIGS. **4** and **5**.
[0032] FIG. **4** is a plan view representing an example of a structure holding the porous body PM provided in the first filter **51**.
[0033] The first filter **51** includes a surrounding body **60** covering the porous body PM. In FIG. **4**, the porous body PM is not exposed and is held to be surrounded by the surrounding body **60**. The surrounding body **60** is formed of a breathable material. The surrounding body **60** includes, for example, a non-woven fabric.
[0034] FIG. **5** is a perspective view of the surrounding body **60** illustrated in FIG. **4**. The surrounding body **60** includes the porous body PM therein, and the peripheral edge is sealed to surround the porous body PM. Airflow in a direction from a first major surface **60***a* of the surrounding body **60** to a second major surface **60***b* opposite to the first major surface **60***a*, or an opposite direction, passes through the porous body PM. The porous body PM adsorbs and removes gas molecules and the like contained in the airflow.
[0035] To reduce resistance when airflow passes through the surrounding body **60**, the first major surface **60***a* or the second major surface **60***b* of the surrounding body **60** is preferably provided in a direction facing the disk **20**.
[0036] The structure for securing the surrounding body **60** provided in the first filter **51** to the base **11** may be, for example, a structure fitting the first filter **51** into a part having a recess or a structure sandwiching the surrounding body **60**.
[0037] The first filter **51** is preferably provided with a mechanism that controls the flow path of the airflow. FIG. **6** illustrates a diagram schematically illustrating the surrounding body **60** provided in the first filter **51** and an airflow guide **62**. The structure of the first filter **51** provided on the side of the disk **20** is illustrated. The first filter **51** includes the surrounding body **60** including the porous body PM and the airflow guide **62** provided downstream of the airflow generated by the rotation of the disk **20**, configured to guide the airflow caused by rotation of the disk **20**. FIG. **6** illustrates an example in which the first major surface **60***a* of the surrounding body **60** is provided in a direction facing the disk **20**. At least a portion of the airflow guide **62** is provided farther from the disk **20** than the surrounding body **60**.
[0038] The airflow guide **62** includes a curved surface **62***c*. At least a portion of the curved surface **62***c* is provided along an arc. Here, the arc may be a circular arc or an elliptical arc. For example, the arc is provided along a curvature circle Cc having a predetermined radius of curvature. A curvature center Oc of the curvature circle Ce is located in the same direction as the direction in which the disk **20** is provided relative to the airflow guide **62**. A curvature center Oc is located near an edge of the disk **20** in a direction that is towards a center of the disk **20**. In other words, the curved surface **62***c* is bent in the same direction as the circular arc, which is the shape of the edge of the circular disk **20**.
[0039] The porous body PM provided in the first filter **51** includes at least a first porous body. Preferably, an average pore diameter of the first porous body is less than an average pore diameter of activated carbon or silica gel. Here, the average pore diameter is measured and averaged for at least two or more pores. The average pore diameter may also be a median or a most frequent value of diameters measured for at least two or more pores. The average diameter of the pores of activated carbon or silica gel is, for example, greater than 1.0 nm.
[0040] The first porous body has pores with an average diameter of, for example, 0.4 nm or more and 1.0 nm or less. To adsorb a gas molecule with a valid diameter of, for example, 0.4 to 0.7 nm, the average diameter of the pores is preferably 0.7 nm or more and 1.0 nm or less. Here, the valid diameter of gas molecules, also known in the art as effective molecular diameter of gas molecules or collision diameter of gas molecules, corresponds to a diameter of the gas molecules assuming a spherical shape that collide with each other, for example.
[0041] The first porous body is a porous body including, for example, zeolite. Hereinafter, when

referred to as zeolite, synthetic zeolite is included. The framework of a zeolite composition is provided as a three-letter designation (framework type code) by the International Zeolite Association. The first porous body is, for example, a zeolite belonging to the framework type code FAU. The first porous body is, for example, an X-type zeolite. The first porous body is, for example, a 13X-type zeolite.
[0042] The first filter **51** may also include activated carbon in addition to the first porous body. The first filter may include a silica gel.
[0043] The first filter **51** may include a second porous body P**2** in addition to the first porous body. The second porous body has pores with an average diameter of, for example, 0.1 nm or more and 0.4 nm or less. Preferably, the average diameter may be 0.2 nm or more and 0.4 nm or less. The second porous body is a porous body including, for example, zeolite. The second porous body is, for example, a zeolite belonging to the framework type code LTA. The second porous body is, for example, an A-type zeolite. The second porous body is, for example, a 3A-type zeolite.
[0044] The operation of the magnetic disk device **100** will be described.
[0045] The disk **20** is rotated by the spindle motor **21**. Based on the signal transmitted from the substrate **40**, the voice coil motor **33** is driven to control the position of the head **31** relative to the rotating disk **20**. The magnetic disk device **100** operates by writing or reading magnetic information at various positions on the disk **20** while changing the positions of the head **31**.
[0046] In response to the rotation of the disk **20**, air flows around the disk **20**. Here, air is not limited to the composition of gases in the atmosphere, but may include a variety of gas molecules. The flow of air on the surface of the disk **20** controls the spacing between the disk **20** and the head **31**.
[0047] Meanwhile, airflow on the side of the disk **20** forms a circular flow. The first filter **51** is located beside the disk **20**, and at least a portion of the air flowing around the disk **20** passes through the first filter **51**. As air passes through the first filter **51**, gas molecules having a valid diameter of a predetermined size are adsorbed by the porous body PM.
[0048] An example of the flow path of the airflow through the first filter **51** will be described with reference to FIG. **6**. As the disk **20** rotates, airflow AF**0** flows into the first filter **51**. At least a portion of the airflow AF**0** flows alongside the airflow guide **62** to form airflow AF**1**. At least a portion of the airflow AF**0**, illustrated as airflow AF**2**, may also flow on the side of the first major surface **60***a* of the surrounding body **60**.
[0049] The direction of the airflow AF**1** changes along the curved surface **62***c* of the airflow guide **62**. The angle at which the airflow AF**1** impinges on the second major surface **60***b* of the surrounding body **60** is controlled by the shape of the curved surface **62***c* of the airflow guide **62**. As the airflow AF**1** is passed through the surrounding body **60** in the direction from the second major surface **60***b* to the first major surface **60***a*, the porous body PM provided in the surrounding body **60** adsorbs gas molecules contained in the airflow.
[0050] Gas molecules with a valid diameter of 1.0 nm or less are known to be difficult to remove in activated carbon and silica gel due to the large average diameter of the pores. The higher the viscosity of the gas molecule, the more difficult it is to remove the gas molecule adhered to the disk or the like. The higher the boiling point of the gas molecule and the harder it is to volatilize, the more difficult it is to separate the gas molecule by heat. When only activated carbon or silica gel is used, there is a risk that the lifetime of the magnetic disk device would be reduced due to gas molecules with high viscosity and boiling point with the valid diameter of 1.0 nm or less.
[0051] Gas molecules with a valid diameter of 1.0 nm or less include, for example, glycol ether or glycol ester. Glycol ether and glycol ester have a greater molecular weight and intermolecular force than, for example, ethanol. It has been found that glycol ether and glycol ester have higher viscosities and boiling points than ethanol and the like, making glycol ether and glycol ester more difficult to remove when attached to the disk or the like.
[0052] Glycol ether includes, for example, ethylene glycol monomethyl ether (2-methoxyethanol),

diethylene glycol monomethyl ether (2-(2-methoxyethoxy)ethanol), ethylene glycol monoethyl ether (2-ethoxyethanol), diethylene glycol monoethyl ether (2-(2-ethoxyethoxy)ethanol), ethylene glycol monobutyl ether (2-butoxyethanol), diethylene glycol monobutyl ether (2-(2-butoxyethoxy)ethanol), propylene glycol monomethyl ether (1-methoxy-2-propanol), or dipropylene glycol monomethyl ether (2-methoxymethylethoxy)propanol). Glycol ether includes molecules having a valid diameter of 0.4 nm or more and 0.7 nm or less.

[0053] Glycol ether may include molecules having a boiling point of 120° C. or more and 240° C. or less. Glycol ether with a higher boiling point than alcohol such as ethanol is included.

[0054] Glycol ether may include molecules with a viscosity of 4 cP or more and 8 cP or less at ambient temperature and pressure. Glycol ether with a higher viscosity than alcohol such as ethanol is included.

[0055] Glycol ester also includes, for example, ethylene glycol monoethyl ether acetate, diethylene glycol monoethyl ether acetate, ethylene glycol monobutyl ether acetate, diethylene glycol monobutyl ether acetate, propylene glycol monomethyl ether acetate, or dipropylene glycol monomethyl ether acetate. Glycol ester includes molecules having a valid diameter of 0.4 nm or more and 0.7 nm or less.

[0056] Glycol ester may include molecules having a boiling point of 120° C. or more and 240° C. or less. Glycol ester with a higher boiling point than alcohol such as ethanol is included.

[0057] Glycol ester may include molecules having a viscosity of 4 cP or more and 8 cP or less at ambient temperature and pressure. Glycol ester with a higher viscosity than alcohol such as ethanol is included.

[0058] According to the magnetic disk device 100 according to the present embodiment, the porous body PM including the first porous body provided in the first filter 51 has excellent performance in adsorbing predetermined contaminants as described later and is capable of extending the lifetime of the magnetic disk device. The first porous body has, for example, pores having an average diameter of 0.4 nm or more and 1.0 nm or less and performs excellently in adsorbing and removing gas molecules having a valid diameter of 1.0 nm or less. For example, the performance in removing glycol ether and glycol ester is excellent. Contamination of the inside of the housing **10** can be reduced and the lifetime of the magnetic disk device **100** can be extended. The average pore diameter of the first porous body is optimized for adsorbing, for example, glycol ether and glycol ester. The upper limit of the the average pore diameter of the first porous body is, for example, 1.0 nm, and such first porous body effectively adsorbs glycol ether and glycol ester having a valid diameter of 1.0 nm or less.

[0059] Glycol ether or glycol ester with a higher boiling point and viscosity than alcohol such as ethanol are easy to adhere to and difficult to remove from the HDD, which may shorten the lifetime of the HDD. According to the magnetic disk device **100** according to the present embodiment, the performance in removing gas molecules having a valid diameter of 1.0 nm or less, including glycol ether and glycol ester, is excellent, and the lifetime of the magnetic disk device **100** can be extended.

[0060] It is preferable that the porous body PM provided in the first filter **51** includes activated carbon in addition to the first porous body. Activated carbon can adsorb molecules of a size that cannot be adsorbed by the first porous body, for example, molecules having a valid diameter greater than 1.0 nm. Accordingly, a wider range of contaminants can be removed to extend the lifetime of the magnetic disk device **100**. Activated carbon may be substituted, for example, with silica gel. Hereinafter, porous body with relatively large pore size, such as activated carbon or silica gel, may be referred to as a third porous body.

[0061] Since the average diameter of the pores of the first porous body is 1.0 nm or less, the efficiency of adsorption can be improved for small gas molecules that are difficult for activated carbon or the like to adsorb as the main target of the first porous body. By inhibiting gas molecules of a size that can be adsorbed by activated carbon and the like from occupying the adsorption site of the first porous body, the first porous body can adsorb more small gas molecules that cannot be adsorbed by activated carbon and the like. Accordingly, gas molecules having a valid diameter of 0.4 nm or more and 0.7 nm or less can be efficiently adsorbed compared to the case where the average diameter of the pores of the first porous body is greater.

[0062] The porous body PM provided in the first filter **51** may further include a second porous body. The second porous body includes, for example, pores having an average diameter of 0.2 nm or more and 0.4 nm or less and has better performance in adsorbing water molecules than the first porous body. Accordingly, the humidity in the housing **10** can be further reduced. The second porous body is superior in adsorbing inorganic molecules such as water molecules, while the first porous body is superior in adsorbing organic molecules such as glycol ethers (small molecules). The average pore diameter of the second porous body is optimized for adsorbing, for example, water molecules.

[0063] By providing the first filter **51** on the side of the disk **20**, gas molecules contained in the airflow generated on the side of the disk **20** as the disk **20** rotates can be efficiently adsorbed. When the spindle motor **21** that rotates the disk **20** is a source of gas molecules, the first filter **51** can be provided near the source to efficiently remove the gas molecules.

[0064] When the source of the gas molecule is identified within the housing **10**, the first filter **51** can be provided near the source of the gas molecule, thereby efficiently removing the gas molecule.

[0065] When the first main surface **60***a* or the second main surface **60***b* of the surrounding body **60** provided in the first filter **51** faces the disk **20**, resistance when airflow passes through can be reduced and the influence on the rotation of the disk **20** can be reduced. By reducing the air resistance by the surrounding body **60**, a decrease in the rotational speed of the disk **20** can be reduced.

[0066] When the first main surface **60***a* or the second main surface **60***b* of the surrounding body **60** faces the disk **20**, the flow path through the porous body PM can be elongated and the efficiency of removing the gas molecules can be improved compared to when the airflow flows at a right angle to the first main surface **60***a* or the second main surface **60***b.*

[0067] The first filter **51** includes the airflow guide **62** illustrated in FIG. **6**, for example, to control the airflow AF**1** to flow at an angle less than **90** degrees to the second major surface **60***b* of the surrounding body **60**. The angle of the airflow AF**1** flowing into the second major surface **60***b* can be controlled by the airflow guide **62**. The shallower the angle to the second main surface **60***b*, the longer the airflow AF**1** can pass through the surrounding body **60**. Accordingly, the efficiency of removing the gas molecules by the porous body PM provided in the surrounding body **60** can be further enhanced.

[0068] The curvature center Oc of the curved surface **62***c* of the airflow guide **62** is located in the same direction as the direction in which the disk **20** is provided with respect to the airflow guide **62**, and the airflow AF**1** is directed toward the second main surface **60***b* while gradually changing the direction thereof, thereby reducing air resistance by the airflow guide **62**.

Second Embodiment

[0069] FIG. **7** is a schematic cross-sectional view illustrating a magnetic disk device **200** according to the second embodiment.

[0070] The magnetic disk device **200** includes the disk **20**, the spindle motor **21**, and the actuator **30** in the housing **10**. The actuator **30** includes the head **31** and the arm **32**.

[0071] The top cover **12** of the housing **10** is provided with the hole **12***h*. The hole **12***h* is not limited to being circular and may be rectangular, elliptical, oval, or the like. The hole **12***h* is not limited to being provided above the disk **20** and may be located above the substrate **40** as also illustrated in FIG. **1**, which illustrates the semiconductor device **1** according to the first embodiment, or above the actuator **30**.

[0072] The inner wall of the housing **10** is provided with a second filter **52** in the area covering the hole **12***h*. The upper surface of the second filter **52** (with the surface facing the base **11** as the lower surface) is in contact with the inner wall of the housing **10**. The second filter **52** is provided with the porous body PM including a first porous body. The porous body PM may further include at least one among activated carbon, silica gel, or a second porous body.

[0073] Next, FIG. **8** is a schematic diagram illustrating an example of the structure of the second filter **52**. The second filter **52** includes a frame **70** including a cavity **70***v*, the porous body PM provided in the cavity **70***v* of the frame **70**, and a ventilation membrane **80** provided on a lower surface of the frame **70** and an opening of the cavity **70***v*.

[0074] The frame **70** includes a hole **70***h* on the upper surface. The hole **70***h* is provided in a position corresponding to the hole **12***h* illustrated in FIG. **7**. The hole **70***h* may be similar in shape to the hole **12***h*. The hole **70***h* may also be the same shape as the hole **12***h*.

[0075] The porous body PM including the first porous body is covered with the frame **70** on the upper side and on the lateral side, and at least a portion of the lower side thereof is covered with the ventilation membrane **80**.

[0076] The second filter **52** may further include a mechanism that controls airflow between the hole **70***h* and the porous body PM.

[0077] The frame **70** includes, for example, plastic resin. The ventilation membrane **80** is a breathable membrane and includes, for example, a non-woven fabric. The porous body PM provided in the second filter **52** may include at least one among activated carbon, silica gel, or a second porous body in addition to the first porous body.

[0078] According to the magnetic disk device **200** according to the present embodiment, a predetermined gas can be sealed in the housing **10** and gas molecules having a valid diameter of 1.0 nm or less can be efficiently removed. For example, when He is sealed in the housing **10**, the inside of the housing **10** is mainly filled by He because He is sufficiently smaller than the average diameter of the pores of the first porous body so that He can pass through without adsorption. On the other hand, the first porous body can remove gas molecules having a larger valid diameter than He, for example, glycol ether and glycol ester. Accordingly, the concentration of gas molecules having a high viscosity and boiling point within the housing **10** can be reduced and the lifetime of the HDD can be extended.

[0079] First, the second filter **52** and the hole **12***h* are described. As an example, the case where He is sealed in the housing will be described. In general, by sealing He, the air resistance due to the gas molecules in the housing can be reduced, and the vertical movement of the head can be reduced.

[0080] When He is injected from the outside of the housing **10** to the inside of the housing **10** through the hole **12***h*, there is a risk that different gas molecules will mix in at the same time. By providing the second filter **52** having the porous body PM including the first porous body in the flow path of the gas molecules to be injected from the outside of the housing **10** to the inside of the housing **10**, it is possible to reduce the incorporation of different gas molecules during He encapsulation.

[0081] The first porous body includes, for example, pores having an average diameter of 0.4 nm or more and 1.0 nm or less, as in the case of the first embodiment. Accordingly, the performance in adsorbing gas molecules having a valid diameter of 1.0 nm or less is excellent. When the average diameter of the pores is 0.7 nm or more and 1.0 nm or less, the incorporation of gas molecules having a valid diameter of 0.4 nm to about 0.7 nm can be further reduced. For example, incorporation of glycol ether and glycol ester can be reduced.

[0082] Even when air is present in the housing **10**, gas molecules adsorbed to the porous body PM of the second filter **52** are discharged from the hole **12***h* to the outside of the housing **10** via the hole **70***h*, and the second filter **52** can further adsorb contaminants.

[0083] Above, an example in which the second filter **52** is provided to contact the top cover **12** was described. The position in which the second filter **52** is provided is not limited to the position in contact with the top cover **12** but can be provided in various positions in contact with the inner wall of the housing **10**. Note that the opening of the cavity **70***v* and the ventilation membrane **80** are

provided on a surface opposite to the surface in contact with the inner wall of the housing **10**.
[0084] Of course, a magnetic disk device may include both the first filter **51** described in the first embodiment and the second filter **52** described in the present embodiment. By including both the first filter **51** and the second filter **52**, the performance of removing the predetermined gas molecules can be further improved.
[0085] According to the semiconductor device of at least one of the first and second embodiments described above, by having a filter structure including the first filter **51** or the second filter **52** having the porous body PM including the first porous body, it is possible to provide a magnetic disk device having excellent performance in removing molecules having a valid diameter of 1.0 nm or less and high viscosity and boiling point, such as glycol ether, and glycol ester. Thereby, the lifetime of the magnetic disk device can be extended.
[0086] Above, embodiments have been described with reference to specific examples. However, the embodiments are not limited to the specific examples. That is, embodiments in which a person skilled in the art has made appropriate changes in design of the specific examples are also encompassed within the scope of the embodiments as long as the embodiments have the features of the embodiments. Each element of each of the foregoing specific examples and arrangement, material, conditions, shape, diameter, and the like thereof are not limited to those illustrated and can be changed as appropriate.
[0087] Each element provided in each of the foregoing embodiments can be combined to the extent technically possible, and combinations thereof are also encompassed within the scope of the embodiments to the extent as long as the features of the embodiments are included. Within the scope of the ideas of the embodiments, it is understood that a person skilled in the art can conceive of various changes and modifications, and such changes and modifications also belong to the scope of the embodiments.
[0088] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

Claims

1. A magnetic disk device comprising: a housing; a disk rotatably provided in the housing; a head provided in the housing and capable of writing and reading information to and from the disk; and a filter structure provided in the housing and including a first porous body with an average pore diameter of 0.4 nm or more and 1.0 nm or less, that is optimized for adsorbing glycol ether.
2. The magnetic disk device according to claim 1, wherein the filter structure further includes a second porous body having an average pore diameter less than the average pore diameter of the first porous body.
3. The magnetic disk device according to claim 2, wherein the second porous body has the average pore diameter of 0.2 nm or more and less than 0.4 nm.
4. The magnetic disk device according to claim 3, wherein the average pore diameter of the second porous body is optimized for adsorbing water molecules.
5. The magnetic disk device according to claim 1, wherein the filter structure further includes a third porous body having an average pore diameter greater than the average pore diameter of the first porous body.
6. The magnetic disk device according to claim 5, wherein the third porous body includes activated carbon or silica gel.

7. The magnetic disk device according to claim 1, wherein the filter structure further includes: a second porous body having an average pore diameter less than the average pore diameter of the first porous body; and a third porous body having an average pore diameter greater than the average pore diameter of the first porous body.
8. The magnetic disk device according to claim 1, wherein the filter structure has an upper surface in contact with an inner wall of the housing and further includes: a frame including a cavity; and a ventilation membrane provided on a lower surface of the frame and covering an opening of the cavity, and the first porous body is provided in the cavity of the frame above the ventilation membrane.
9. The magnetic disk device according to claim 1, wherein the first porous body includes pores adsorbing a glycol ether with a valid diameter of 0.4 nm or more and 0.7 nm or less.
10. The magnetic disk device according to claim 1, wherein the first porous body includes pores adsorbing a glycol ether including at least one type of molecule selected from ethylene glycol monomethyl ether, diethylene glycol monomethyl ether, ethylene glycol monoethyl ether, diethylene glycol monoethyl ether, ethylene glycol monobutyl ether, diethylene glycol monobutyl ether, diethylene glycol monobutyl ether, propylene glycol monomethyl ether, or dipropylene glycol monomethyl ether.
11. A magnetic disk device comprising: a housing; a disk rotatably provided in the housing; a head provided in the housing and capable of writing and reading information to and from the disk; and a filter structure provided in the housing and including a first porous body and a second porous body having an average pore diameter less than the average pore diameter of the first porous body, the filter structure including pores with an average pore diameter of 0.4 nm or more and 1.0 nm or less.
12. The magnetic disk device according to claim 11, wherein the first porous body has an average pore diameter of 0.4 nm or more and 1.0 nm or less.
13. The magnetic disk device according to claim 11, wherein the filter structure further includes a third porous body having an average pore diameter greater than the average pore diameter of the first porous body.
14. The magnetic disk device according to claim 11, wherein the second porous body has an average pore diameter of 0.4 nm or more and 1.0 nm or less.
15. A magnetic disk device comprising: a housing; a disk rotatably provided in the housing; a head provided in the housing and capable of writing and reading information to and from the disk; a voice coil motor provided in the housing and driving the head; and a filter structure provided in the housing and located closer to the disk than to voice coil motor.
16. The magnetic disk device according to claim 15, wherein the filter structure is located between the voice coil motor and an edge of the disk.
17. The magnetic disk device according to claim 15, wherein the filter structure is located on an opposite side of the disk relative to the voice coil motor.
18. The magnetic disk device according to claim 15, wherein the filter structure further includes an airflow guide configured to guide an airflow caused by rotation of the disk.
19. The magnetic disk device according to claim 18, wherein the airflow guide is provided upstream of the airflow flowing into the filter structure.
20. The magnetic disk device according to claim 18. wherein the airflow guide further includes a curved surface having a curvature center located near an edge of the disk in a direction that is towards a center of the disk.